

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0837

Virginia Tech Power Plant 138 kV
November 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0837. The Interconnection Customer proposes a 61.9 MW solar photovoltaic facility interconnecting at the Virginia Tech Power Plant 138 kV substation in the Virginia area. The request was received on July 27, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 338 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$724,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NEW LONDON / DECLO 138kV	BATH COUNTY / SHERBURNE 345kV	116.0	307	8.6
CHATOM / PURDY 138kV	FRITO LAY / DPC PRAIRIE DU CHIEN 138kV L	110.9	580	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NEW LONDON / DECLO 138kV	Rebuild line with higher-capacity conductor	\$41,002,000	2% (\$650,000)
PURDY / CHATOM 138kV	Rebuild line with higher-capacity conductor	\$6,645,000	1% (\$74,000)
	Total allocated to project		\$724,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.